Offshoring Production while Offshoring Pollution?

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CES 16-09 **February, 2016**

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Abstract

We combine international trade data from the U.S. Census Bureau with Toxics Release Inventory data from the Environmental Protection Agency to investigate the relationship between U.S. firms' imports from low-wage countries (LWCs) and toxic emissions by their domestic plants. We find that plants release less toxic emissions on American soil when their parent firm imports more from LWCs. According to our estimates, when a plant's parent firm increases its share of imports from LWCs by 10 percentage points, the plant's toxic emissions on American soil decrease by about 4%. These effects are stronger for plants facing greater institutional pressures on environmental performance, such as plants located in dirtier U.S. counties. In addition, goods imported by U.S. firms from LWCs are in more pollution-intensive industries than goods imported from the rest of the world. Taken together, our results provide the first large-sample empirical evidence that U.S. firms offshore both production and pollution to the developing world.

Keyword*:* environment, offshore, institutions

[*] The research in this paper was conducted while both authors were Special Sworn Status researchers of the U.S. Census Bureau at the Michigan Census Research Data Center (RDC). Support for this research at the Michigan Census RDC from NSF (ITR-0427889) is gratefully acknowledged. Any opinions and conclusions expressed herein are those of the authors and do not necessarily represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. We thank Nathan Wilson for helpful comments and Randy Becker for information about the PACE surveys.

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1. Introduction

Over the past few decades, the United States has enjoyed a significant reduction of pollution in its environment (Levinson 2009). Much of this reduction has been attributed to strict environmental regulations (Chay and Greenstone 2003, Shapiro and Walker 2014) and informal institutional pressure (Marquis and Toffel 2014). Regulators such as the United States Environmental Protection Agency (EPA) frequently report on the dirtiest plants and counties based on pollutant emissions; such information compels local governments and communities to force polluting plants to either reduce emissions or to pay a hefty penalty (Forbes 2012). Investors are demanding a higher return on their capital from companies with environmental issues and have been pressing companies to report material information about not only financial performance but also social sustainability performance, including information about companies' pollutant emission levels and risks (Chava 2014, Reid and Toffel 2009, The Wall Street Journal 2015). Complying with strict legal and informal institutional requirements, however, imposes significant operating costs (Blass et al. 2014).

Prior studies reveal a few environmental strategies to cope with greater environmental demands. First, firms can choose to comply with environmental standards by investing in waste treatment and prevention (King and Lenox 2002). Second, firms can seek to lower the cost of compliance by innovating products and processes (King and Lenox 2001, Porter and Van der Linde 1995), innovating business models such as switching from manufacturing to servicizing (Bellos et al. 2013, Guajardo et al. 2015, Toffel 2002) or from selling to leasing (Agrawal et al. 2012). Finally, firms can try to avoid the cost of compliance by greenwashing (Kim and Lyon 2015), or presenting an image of environmental responsibility without actual implementation. These strategies, however, can be constrained by firm capabilities (Barnett and Salomon 2012, Berchicci et al. 2012) or societal scrutiny (Marquis and Toffel 2014).

In this paper, we explore yet another environmental strategy by which firms reconfigure their domestic product portfolio by leveraging the global production system. With globalization, companies have increasingly dispersed their production and sourcing networks to take advantage of locational advantages and global production scale, causing "the increasing interconnectedness of production processes in a vertical trading chain that stretches across many countries" (Hummels et al. 2001: 76). We argue that firms might separate their production sites—that is, they can redesign their supply chain to shift their domestic production to cleaner segments,

thereby complying with the U.S. standards, and produce in or import from poor or low-wage countries (LWCs) products that require more-polluting production process. Even though the comparative advantage of LWCs should attract more labor-intensive industries rather than the more pollution-intensive capital-intensive industries (Cole and Elliott 2005), LWCs also have lax environmental standards and poor environmental regulatory quality (Esty and Porter 2002), which can attract foreign firms based on environmental rather than labor considerations.

We linked firm-level imports and plant-level production statistics maintained by the U.S. Census Bureau to plant-level toxics emissions information from the EPA's Toxics Release Inventory (TRI) database, a database that has been widely used in prior studies of environmental performance (Doshi et al. 2013, King and Lenox 2000, King and Lenox 2002, King and Shaver 2001, Lee and Lounsbury 2015, Reid and Toffel 2009). We distinguished between imports from LWCs and imports from the rest of the world. Using firm-level trade and plant-level production and pollution data offers several benefits. First, it fills in a gap in prior research about the impact of globalization on the environment, which is mostly at the country-, state-, and industry-level due to data constraints. Second, it allows us to control for multiple alternative explanations for environmental performance such as scale of production, capital vs. labor intensity, and skilled vs. unskilled labor requirement at the plant level. Finally, using plant-level regression allows us examine the impact of different local institutions on individual plants within the same firm.

Our results suggest that domestic plants pollute less on American soil as their parent firm imports more from LWCs: When a plant's parent firm increases its share of imports from LWCs by 10 percentage points, the plant's toxic emissions on American soil fall by 4%–6%. Further analyses show that imports from China play a large role in reducing toxic emissions in the United States, but our results continue to hold when we consider imports from non-China LWCs. Consistent with a potential substitution between domestic and offshored pollution-intensive production, we also found that goods imported by U.S. firms from LWCs are in more "pollution intensive" industries—industries that emit more pollutants per dollar of output in the United States—than goods imported from the rest of the world. In addition, U.S. plants produce more goods in less pollution-intensive industries as their parent firm imports more from LWCs.

In addition, we examined the impact of local institutional pressure on offshoring. The EPA publishes an annual list of the most polluted counties in the United States. Drawing attention to these "dirtiest counties" invites an aggressive response from the media and local communities

and implies a higher threat of inspection and penalty from local governments. We therefore expect that plants located in dirty counties will offshore more than plants located in cleaner counties. Our results show that indeed plants facing greater institutional pressures on environmental performance, such as plants located in dirtier U.S. counties, reduce their domestic pollution even more as their parent firm increases imports from LWCs. When a parent firm increases its share of imports from LWCs by 10 percentage points, a plant in a clean county will reduce its total toxic emissions by 3%. In contrast, a plant located in a dirty county will reduce its total toxic emissions by an additional 4%. These results are robust after controlling for alternative explanations such as offshoring based on labor costs.

This paper's main contribution is to introduce a global dimension to firms' strategic response to environmental standards. While the impact of global standards, such as the ISO9000 and ISO14000 quality certifications, on local states and organizations has been studied (Albuquerque et al. 2007, Guler et al. 2002, Polillo and Guillén 2005), the reverse scenario, in which firms respond to local standards using globalization has not. Our finding that firms could switch to cleaner segments in U.S. but source from LWCs products that require pollution intensive production processes suggests that firms could respond to institutional pressure by compliance at the local level but avoidance at the global level. While we do not offer any moral endorsement of such "jurisdiction shopping" strategy (Ahuja and Yayavaram 2011), we demonstrate that such strategy is conceptually possible based on institution theories and is adopted by firms in practice.

In addition, this paper provides the first micro-level empirical evidence that the "green shift" of U.S. manufacturing corresponds with a "brown shift" of imports from poor countries. This is consistent with the Pollution Haven Hypothesis (PHH) in trade theory that, "Liberalized trade in goods will lead to the relocation of pollution intensive production from high income and stringent environmental regulation countries to low income and lax environmental regulation countries" (Taylor 2005). For example, a recent study in China using atmospheric modeling found that 17%–36% of four major anthropogenic air pollutants (SO2, NOx, CO, and black carbon) emitted in that country are associated with the production of goods for export, and that about 21% of export-related emissions are attributable to goods destined for the United States (Lin et al. 2014). A natural question is what the U.S. pollution level would have been had those goods been produced in the United States rather than in China or other poor countries. However,

most of the prior research on the environmental impact of globalization has relied on aggregate country-, state-, or industry-level information (Antweiler et al. 2001, Ederington et al. 2004, Grossman and Krueger 1995, Hanna 2010, Levinson 2009, Levinson 2010), which partly explain some contradictory results in this research. In addition, the level of analysis in existing studies cannot distinguish between extensive adjustments (e.g., firm/plant entry and exit) and intensive/strategic adjustments (e.g., product portfolio reconfiguration) at the firm level. The current paper fills in this gap by specifically focusing on the intensive adjustments.

Our paper also has significant policy implications. It implies that strict domestic environmental standards could mitigate negative externality in one institution but exacerbate negative externality in far-flung institutions due to globalization. With the recent public debates around re- or near-shoring vs. off-shoring, the impact of institutional constraints on the configuration of global production networks becomes increasingly salient. While conventional wisdom assumes that pollution is local, recent research proves it is becoming a global concern. For instance, studies show that pollution from China contributes a significant portion of the sulfate concentrates found over the western United States (Lin, et al. 2014). Our empirical findings lend further credence to policy makers' assertions that there should be more coordination between international trade agreements and domestic environmental regulations (Keller and Levinson 2002).

2. Theory Development

Institutions are humanly devised constraints that structure human interactions; they include formal rules as well as informal norms of behavior and conventions (North 1990). Institutions can be both supportive and detrimental to organizations. On the one hand, institutions provide information, clarity of property rights, and "rules of the game". Information enables better monitoring and measuring of effort and performance, and clear property rights and "rules of the game" facilitate enforcement (Alchian and Demsetz 1972, Coase 1937, Demsetz 1967). Accordingly, weak institutions raise transaction costs, and strong institutions promote economic growth (North 1990). On the other hand, institutions can be developed independent of efficiency and diffused through coercive, mimetic, and normative processes (DiMaggio and Powell 1983, Meyer and Rowan 1977). In particular, the diffusion of global institutions impacts national and

state governments (Henisz et al. 2005, Polillo and Guillén 2005, Schneper and Guillén 2004), capital markets (Soleimani et al. 2014), and organization practices (Guler, et al. 2002). Because organizations require legitimacy to survive and thrive (Weber 1924), institutions exert powerful pressure on firms to conform (Orru et al. 1991).

Institutions vary not only by their strength and benefits/costs to organizations, but also by time and space. This variation creates an opportunity for organizations to arbitrage across institutional boundaries. In a closed social system firms might choose to comply with institutional demands despite of the high cost of compliance. For example, "compliance with the variety of procedures specified by the Environmental Protection Agency elevates its legitimacy and protects it from public criticism and the financial penalties of noncompliance" (Oliver 1991: 153). However, globalization relaxes the constraint of local institutions, which allows firms to arbitrage between varying institutional constraints, effectively putting "sovereignty at bay" (Kobrin 2001, Vernon 1971). For example, firms can circumvent trade barriers through foreign direct investment (Caves 1996) or evade tax liability by relocating operations to low-tax countries (Desai et al. 2004).

Arbitraging between environmental standards

The impact of strict environmental standards on U.S. firms has been controversial in policy as well as in academic debate. Some scholars argue that shareholders benefit from firms' green initiatives because socially responsible firms can deter stakeholder activism (Reinhardt 1999), attract consumers (Casadesus-Masanell et al. 2009, Elfenbein and McManus 2010, Sen and Bhattacharya 2001, Servaes and Tamayo 2013) and motivate productive employees (Flammer 2015, Greening and Turban 2000, Prendergast 2007, Tonin and Vlassopoulos 2015), thereby enhancing shareholder value (Eccles et al. 2014, Porter and Kramer 2011). Other scholars have argued that strict environmental standards raise firms' production costs and weaken U.S. firms' competitive position in international markets. In particular, some economists have blamed strict environmental standards in the U.S. for declining manufacturing productivity (Greenstone et al. 2012), plant closures (Becker and Henderson 2000, Henderson 1996), losses of American jobs (Greenstone 2002), and falling wages for U.S. workers (Walker 2013). Some even point out that "the decline in U.S. manufacturing employment from 18 million (25.3 percent of total U.S. employment) in 1970 to 12 million (9.0 percent of total employment) in 2012 mirrors the

introduction and expansion of U.S. environmental policy" (Greenstone, et al. 2012: pg. 1). The concern over "job killing" environmental regulations in the United States is one major obstacle to trade agreements such as NAFTA (Greenstone, et al. 2012).

The benefit of marketing environmentally friendly brands to stakeholders versus the cost of pollution prevention and treatment presents a dilemma for firms. There are two obvious responses to the strict environmental standards at home: compliance or avoidance. We argue that firms can integrate both strategies by separating their production sites—that is, they can redesign their supply chain to locate more-polluting production process in LWCs, thereby escaping U.S. pollution standards, and shift their domestic production to cleaner segments, thereby complying with the U.S. standards. Of course, firms can adopt strategies that are less dramatic than offshoring. They can, for example, compromise by redesigning their production process and upgrading their products to reduce pollution (King and Lenox 2001, Oliver 1991, Porter and Van der Linde 1995). Not all firms are capable of such innovation (Berchicci, et al. 2012), however, especially given the short-term orientation of capital markets. Firms may also engage in greenwashing, in which they manipulate public perception of their environmental performance by selectively disclosing positive environmental actions and concealing negative ones (Kim and Lyon 2015). However, such strategic "cheap talk" is not always feasible, particularly in locations where there is heavy regulatory surveillance (Short and Toffel 2010), where social activism is easy to mobilize (e.g., through media), and where more information and societal scrutiny create more normative pressure to align policies and practices (Bromley and Powell 2012, Marquis and Toffel 2014).

There is some evidence that firms arbitrage between environmental standards. For example, it has been found that, in response to the Clean Air Act Amendments (CAAA), firms use air pollution abatement techniques to remove pollutants from the air but release them into water bodies, landfills, or the ground (Greenstone 2003). In addition, firms operating multiple facilities across multiple jurisdictions in the United States generate more waste (King and Shaver 2001). Furthermore, firms invest less in U.S. states where the pollution abatement costs are expected to be high (Keller and Levinson 2002), and firms in U.S. counties where environmental regulations are expected to be more stringent invest more outside the United States (Hanna 2010). In general, firms from rich countries have been criticized for pitting poor countries against one another to create a "race to the bottom" in terms of labor and environmental standards (Gladwin et al. 1995)

and offshoring pollution-intensive production to low-wage countries. Consequently, *we expect U.S. plants to reduce their domestic pollution as their parent firm increases imports from LWCs.*

Local institutional pressure and global response

Both public regulatory agencies and private-sector actors such as the news media can exert significant institutional pressure on firms for compliance (Deephouse 1996, Hiatt et al. forthcoming). Voluntary information disclosure programs such as the EPA's TRI program have become the third most prevalent policy tool for disciplining firms' social behavior, following command and control and market-based regulations (Tietenberg 1998). The most significant impact of these programs is that they provide stakeholders, including local communities, with information for making decisions and for exerting pressure on non-conforming firms (Konar and Cohen 1997). Even though some of these social activists might not have a formal contractual relationship with the firm (and are hence referred "secondary" stakeholders), they can engage in protests, civil suits, and letter-writing campaigns to exert pressure on firms. They can force firms to meet their demands by either imposing direct operational costs such as legal fees, public relations expenses, and managerial attention or by threatening the firm's reputation amongst its customers, employees, regulators, and shareholders (Eesley and Lenox 2006). Both anecdotal and empirical evidence has shown that informed neighborhoods can apply pressure to induce firms to cut emissions or relocate production (Khanna et al. 1998, Maxwell et al. 2000, Pargal and Wheeler 1995).

Firms' incentive to arbitrage between institutions depends on the pressure of local institutions on the one hand and the cost of arbitrage on the other. While local stakeholders in the United States have become more powerful in part by leveraging information to raise the social consequence of firm behavior, communities in poor countries have less power against their states due to both a lack of information and a lack of property rights. They might also have different incentives, as their basic economic needs have yet to be met. This results in more lax and lenient environmental standards in those countries. When local communities in the United States can be expected to react more aggressively to poor environmental performance, U.S. firms become more likely to offshore. *We therefore expect U.S. plants facing stronger local institutional pressure to reduce their domestic pollution even more as their parent firm increases imports from LWCs.*

3. Globalization and Environment

Consistent with prior statistics (Levinson 2009), Figure 1 shows that the emissions of major air pollutants by U.S. manufacturers fell by more than half between 1992 and 2009, despite the significant growth in real U.S. manufacturing output. While strict environmental regulations and informal institutional pressure for environmental performance in the United States have been widely praised for such improvements, they nevertheless impose significant operating costs on American firms (Blass, et al. 2014). This might explain the lack of conclusive evidence on the payoff of a "green" strategy. On the one hand, a recent survey indicated that more than half of consumers across 60 countries are willing to pay more for products and services with a positive social and environmental impact (Nielsen 2014), and some academic research shows that firms with strong environmental management enjoy greater accounting returns (Hart and Ahuja 1996, Nehrt 1996, Russo and Fouts 1997) or higher financial returns (Dowell et al. 2000, Klassen and McLaughlin 1996) or both (King and Lenox 2002). On the other hand, some economists have blamed strict U.S. environmental regulations for declining manufacturing productivity (Greenstone, et al. 2012), plant closures (Becker and Henderson 2000, Henderson 1996), losses of American jobs (Greenstone 2002), and falling wages for U.S. workers (Walker 2013). Balancing the benefits and costs of complying with strict environmental standards therefore presents a serious challenge for firms.

Insert Figure 1 about here

Figure 2 shows that while imports from LWCs have been small historically, they have increased substantially in recent years as trade barriers have been removed. Between 1992 and 2009, when the real value of total U.S. imports more than doubled, the real value of imports from LWCs grew more than seven-fold. Consequently, the share of total U.S. imports from LWCs in this period rose from 7% to about 23%. At the national level, the increasing share of imports from LWCs in Figure 2 corresponds to the decreasing air pollution in Figure 1.

Insert Figure 2 about here

Whereas the annual cost of complying with environmental standards in the Unites States amounts to hundreds of billions of dollars and more than two percent of GDP, most less developed countries spend only a fraction of one percent (Jaffe et al. 1995) and have not been able to adopt strict environmental standards for fear of hurting economic growth (The Economist 1998). Such differences create a unique arbitrage opportunity for U.S. firms, who may offshore some polluting production processes to poor countries with lax environmental standards and refocus their U.S. plants on a "cleaner" segment of the value chain. At the industry level, Figure 3 shows that between 1992 and 2009, those industries that had a greater increase in the share of imports from LWCs also experienced a greater reduction in air pollutant emissions from U.S. plants. For example, industries that experienced the greatest increase in imports from LWCs were in the sectors of printing (SIC 278), apparel and textile (SICs 235 and 238), rubber and plastics (SIC302), and furniture (SICs 251, 254, and 259); all experienced some of the largest drops in air pollution emissions. In contrast, industries that had the least increase in imports from LWCs due to transportation costs or trade barriers, including industries in the sectors of food (SIC 208), stone, clay, and glass products (SIC 325), and tobacco products (SIC 214), experienced the least improvement in air pollution emissions. These patterns suggest a potential substitution effect between LWC imports and domestic emissions.

Insert Figure 3 about here

The EPA's Toxics Release Inventory (TRI) program was the first large-scale initiative to track facility-level pollution emissions in the United States. Introduced by the Emergency Planning and Community Right to Know Act (EPCRA) in 1986, the TRI program intends to provide public environmental information and to affect firm behavior indirectly through consumer, public, or community pressure (Konar and Cohen 1997). The TRI public database

contains approximately 80,000 facility-chemical reports from more than 20,000 different facilities. The EPA requires facilities that emit more than 25,000 pounds or handle more than 10,000 pounds of any of the 600+ designated toxic chemicals to self-report emissions data for use in a publicly available database. One recent study found that, on average, the opening of a manufacturing plant that appears in the TRI database increases the likelihood of low-birth weight by roughly 3% for babies born within a one-mile radius during each year of the plant's operation (Currie et al. 2015).

Although the TRI database is based on self-reported information, there are a few mechanisms to ensure its accuracy. First, each EPA region has a TRI enforcement program that conducts, on an annual basis, a limited number of data quality inspections (of reporting facilities) and non-reporting inspections (of facilities that are in TRI industries but did not report). Violations, whether stemming from late reporting, failure to report, or data quality issues, can lead to penalties of $25,000 per day, per chemical, or per violation, and may be subject to criminal charges. Second, a complementary self-reporting program, EPA's Audit Policy program, launched in 1995, reduces or waives certain penalties for environmental violations that are voluntarily disclosed to the government by regulated entities. Such self-reports are found to help regulators identify self-policing firms and reallocate enforcement resources to firms that do not self-report (Toffel and Short 2011). These enforcement mechanisms give firms incentive to self-report.

The TRI database has been used by a large body of stakeholders including the government, investors, potential employees, industry, media, and general public to obtain information on environmental risks and performance at the plant level. Partly in response to that pressure, plants that report to TRI dropped their total pollutant emissions by about 60% between 1988 and 2005. The TRI database has also become one of the most widely accessed databases providing comparative data on environmental performance across facilities and over time. It has been used by environmental researchers for studies addressing a wide range of topics. Prior research shows that residential house prices in heavily polluted areas declined after the TRI database was published (Oberholzer-Gee and Mitsunari 2006). Both public media and the stock market respond negatively when a firm reports higher emissions in the TRI (Hamilton 1995). Firms that have experienced the deepest stock price declines in response to their TRI reports have subsequently reduced emissions more than their industry peers (Konar and Cohen 1997). This

evidence suggests that both the public and firms take TRI information seriously in their investment/operation decisions. In fact, the EPA concluded from its 2000 assessment that "national publication of the TRI data by the government, followed by analysis by citizens' groups and the news media, led to action by industry to reduce emissions" (EPA 2000, Oberholzer-Gee and Mitsunari 2006).

In addition to publicizing plant-level emissions information, the EPA also reports annually on aggregate pollution at the county level and provides lists of the "dirtiest" counties. Such lists are often picked up by news agencies and local communities, which lobby consumers and compel local governments to force polluting plants to either reduce emissions or pay a hefty penalty. For example, Fresno California was named as the dirtiest county by EPA's annual report in 2012. In the same year, "California held its first auction for carbon allowances under the state's new cap-and-trade law that will force any company that emits more than 25,000 tons of carbon dioxide a year to reduce its emissions, or pay the price for permits" (Forbes 2012). Other examples are given by Powers (2013). After Calhoun County, Texas, was listed as dirtiest, local communities organized various awareness programs to inform the public about local pollution. Under public pressure, Alcoa had to commit to aggressive pollution reduction initiatives at two local plants. Similarly, when Butler County, Pennsylvania, was identified among the dirtiest counties, local communities successfully pressured the state to restrict nitrate emissions of a major steel plant before the plant was allowed to release waste into the Connoquenessing Creek. The anecdotal evidence suggests that plants located in "dirty" counties face much greater public pressure to reduce toxic releases than plants located in cleaner counties.

4. Empirical Design

4.1. Data and variables

We constructed our samples, which extend from 1992 to 2009, from several sources. First, we used the plant-level toxic emissions data published in the EPA's TRI database. A plant—called a "facility" in the TRI and an "establishment" in Census Bureau terminology —is a physical location where economic activity takes place. A firm can own one or multiple plants.

Certain pollutants have more-severe health consequences than others (Dockery et al. 1993, Pope et al. 2002). Toffel and Marshall (2004) compare 13 methods of aggregating chemical-specific release data to the plant level and recommend EPA's Risk-Screening Environmental

Indicators (RSEI) model as the most comprehensive model for estimating the impacts of toxic releases on local residents' human health. The RSEI model estimates the toxicity weight for each type of air and water release or transfer of each chemical. Each chemical's toxicity weight is estimated based on its toxicity, its fate and transport through the environment after its release, the route (inhalation and oral) and extent of human exposure, and its single most sensitive adverse effect (cancer and non-cancer) (EPA 2012). The 2010 TRI contains 611 chemicals and chemical categories, 433 of which have toxicity weights; they account for more than 99% of the reported pounds for all on-site releases in 2010 (EPA 2012).

We used the TRI database and the RSEI model to construct four variables. The first variable gauges plant-level toxic emissions in the United States. We followed recent studies using the RSEI model to define toxic emissions from a plant as its all-media release of designated toxic chemicals, multiplied by the RSEI toxicity weight for each chemical; emissions to air are weighted using inhalation toxicity and emissions to other media are weighted using oral toxicity (Gamper-Rabindran 2006). The second variable calculates the pollution intensity for each 4-digit SIC industry. We summed the RSEI-based toxic emissions of all plants in each industry in 1992 and divided the sum by the total output from that industry in 1992, based on the NBER-CES Manufacturing Industry Database (Bartelsman and Gray 1996), to derive a pollution intensity measure for that industry. We then summed a firm's import value in each industry, weighted by the industry's pollution intensity, to derive the overall pollution intensity of a firm's imports. We constructed the third variable with a similar methodology: We summed a plant's output in each industry, weighted by the industry's pollution intensity, to derive the overall pollution intensity of a plant's output. The fourth variable identifies the 100 dirtiest counties. We aggregated the RSEI-weighted plant-level toxic emissions to the county level, based on annual emissions. We then selected the 100 "dirtiest" counties based on the ranking of total emissions. As an alternative measure, we also selected the 100 "dirtiest" counties based on the ranking of total emissions the initial year (1992); results were similar.

Secondly, we used plant-level micro data from the U.S. Census Bureau to control for other factors that might influence pollution emissions, such as plant size (total shipment value of output), capital expenditure, and workers' skill intensity (non-production worker salary as a percentage of total workforce salary). The Census micro data on manufacturing plants include the Census of Manufactures (CM) and the Annual Survey of Manufactures (ASM). CM data are

collected during the economic census, which takes place in years ending in 2 and 7, and covers approximately 350,000 manufacturing plants each time. The ASM typically samples about 60,000 plants in non-census years. All plants with more than 250 employees and all plants of large firms are included by design. Some 40,000 other plants are selected with a probability proportional to a composite measure of their size. Once a plant is surveyed, the ASM continues surveying it to form a five-year panel. We linked the Census and TRI datasets by using the existing bridge files maintained by the Census Bureau and also by manually matching plant names and addresses.

Thirdly, we used the Longitudinal Firm Trade Transaction Database (LFTTD) for information on imports. The database covers all transactions of goods that crossed U.S. borders. For each transaction, the database contains a firm identifier and pertinent details of the transaction, such as the date, value, and destination (or originating) country of the shipment, as well as a 10-digit Harmonized System (HS) classification code of the product category. We followed Pierce and Schott (2012) by linking the ten-digit Harmonized System (HS) classifications to the 1987 version of four-digit SICs. To identify LWCs, we relied on the list provided by Bernard et al. (Bernard et al. 2006), who classify a country as an LWC if its average annual GDP per capita was less than 5% of the U.S. annual GDP per capita from 1972 to 1992. A list of LWCs is provided in Table 1. China, India, and most African countries are on the list. We calculated a firm's LWC Import Share as the percentage of its total imports that originated from LWCs. LWC Import Share has risen substantially according to our sample statistics: from 7% in 1992 to about 23% in 2009.

Insert Table 1 about here

Our main sample is the intersection of plants' pollution data from TRI, plants' operating information from ASM and CM, and firm's imports from LFTTD. This sample contains 17,773 plants of 7,115 U.S. parent firms for a total of 136,574 plant-year observations between 1992 and 2009. Table 2 provides summary statistics for this sample. An average importing firm sources 16% of its manufacturing imports from LWCs, slightly higher than the national average of 15%. The

skill intensity variable has a mean of 0.35; that is, non-production workers' salaries account for about 35% of an average plant's total salaries. The sample plants are relatively large: A typical plant has about 418 employees and manufactures a total value of $175 million of output. A simple calculation reveals that our importing firm sample accounts for about 90% of all manufacturing plants' total toxic emissions in the sample period.

Insert Table 2 about here

Table 3 reports the pairwise correlation among our key variables of interest. As shown in Table 3, a plant's toxic emission is negatively correlated with its parent firm's share of imports from LWCs, but positively related to the plant's size, capital expenditures, and total volume of imports. These simple correlation statistics foreshadow our subsequent multivariate regression results.

Insert Table 3 about here

4.2. Specifications

Our main analyses focus on the implications of parent firm imports on subsidiary plants' domestic pollution reduction. We used the following specification to estimate toxic emissions at the plant level:

$$\text{Toxic Emission}_{ijt} = \alpha_i + \alpha_t + \beta\text{LWC Import Share}_{jt} + X_{ijt} + \varepsilon_{ijt}, \quad [1]$$

where $Toxic\ Emissions_{ijt}$ is measured in two ways: (1) the logarithm of the total toxicity-weighted emissions of plant *i* of parent firm *j* in year *t*, and (2) the logarithm of plant *i*'s total toxicity-weighted emissions scaled by its output, in year *t*. $\text{LWC Import Share}_{jt}$ is parent firm *j*'s share of imports from LWCs. We control for several plant-level characteristics that are known to

affect a plant's toxic emission, such as the plant's scale (the logarithm of its sales),[1] the logarithm of its capital expenditures, and the skill intensity (share of non-production workers over the entire workforce), which can proxy for the characteristics of a plant's production technology. We also control for the parent firm's total imports to examine how imports in general affect a plant's pollution. Standard errors are clustered at the firm level to allow for group effects within firms, since imports only vary at the firm level. In addition to OLS regression with plant fixed effects, we also used an instrumental variable approach to sharpen the identification. This is explained in detail in the next section.

We then expand the main specification in Equation 1 in two ways. First, we investigate additional dependent variables. They include the pollution intensity of industries in which U.S. firms import goods from LWCs and the pollution intensity of industries in which U.S. plants produce goods domestically. These additional variables help to explore a complete picture of substitution between imports from LWCs and domestic pollution. Second, we include an interaction term between LWC imports and local institutional pressure to explore the additional impact of local institutional pressure.

5. Analyses and Results

5.1. Imports from LWCs and domestic pollution

Table 4 reports our main results based on Equation 1. All columns include year fixed effects. In addition, column (1) includes year fixed effects, and columns (2)–(3) include industry and plant fixed effects to control for changes in industry-specific and plant-specific technology and reductions in trade costs over time.

Insert Table 4 about here

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[1] Using alternative measure of scale, such employment, returns similar results.

The results are qualitatively similar across all columns. They show that a firm's LWC Import Share has a significantly negative impact on its domestic plants' toxic emissions, as well as on its toxic emissions per dollar of shipment. The economic effect of the point estimates is considerable. For instance, the coefficient of –0.559 in column (2) implies that a 10% increase in a plant's parent firm's LWC Import Share lowers the plant's toxic emissions by about 5.6%. Our results imply that, over the 18-year sample period, when the economy-wide share of import from LWCs grew by 16 percentage points, a plant would reduce its toxic emission by about 9%, a reduction that accounts for around 15% of the total drop in U.S. toxic emissions in the period. Coefficients on other explanatory variables are consistent with our expectation. In general, larger plants, plants with larger capital expenditures, and plants with a larger proportion of production workers tend to produce greater toxic emissions. Finally, total import does not have a statistically significant impact on toxic emissions and its inclusion does not qualitatively change the coefficients of LWC Import Share.

Although the regressions in Table 4 control for various determinants of pollution emission and plant fixed effects that remove time-invariant plant-level factors that can influence emission, they do not capture all the unobservable factors that may drive both reduction in pollution emissions and imports from LWCs. Two obvious candidates for such unobservable factors are regulation and technology. Firms may be reducing pollution in the United States due to more strict environmental regulations or because they have developed new technologies to reduce pollution; at the same time, they may be increasing imports from LWCs due to reduction in U.S. tariffs for imports. For example, prior research shows that firms that offshore are more innovative, productive firms (Bernard et al. 2007). Technologically advanced firms might also be more likely to adopt clean technologies. These unobservable factors might cause a spurious relationship between imports and toxic emissions.

To address the problem of omitted variables, we first used a longitudinal rather than cross-sectional analysis based on a panel of plant-level dataset. In addition, we included plant and industry-year fixed effects to account for unobservable time-invariant factors at the plant-level (such as innovativeness and technological capability) and industry-specific yearly events that would affect pollution (such as industry-specific regulatory change or technology breakthrough).

5.2. Substitution between domestic pollution and offshoring

While Table 4 suggests that imports from LWCs reduce pollution emissions by U.S. plants, it does not prove a substitution effect between pollution in U.S. and pollution in LWCs. An alternative explanation could be that the production of environmentally friendly products is cheaper in LWCs than in U.S., and the availability of such cheaper, "green" products encourages more American consumers to switch to these products, which further encourages American firms to produce them, resulting in better environmental performance at U.S. plants. To examine if there was a substitution effect between domestic pollution and offshoring, we first estimated the relationship between a firm's LWC Import Share and the pollution content and intensity of its imported goods. Results are presented in Table 5. The dependent variable in the first two columns, Toxic imports, is the total value of the firm's imports across all manufacturing industries, weighted by industry-level pollution intensities based on the 1992 TRI, as described earlier. The dependent variable in the last two columns represents the pollution intensity of the firm's imports, or its "toxic import" scaled by its total imports. The coefficients show that the dirtiness of a firm's imports is positively and significantly related to its LWC Import Share. The economic significance of the point estimate is sizeable. Coefficients in column (2) imply that a 10-percentage-point increase in a firm's LWC Import Share is associated with a 3.6% increase in the amount of its "toxic imports". As a robustness check, we constructed an alternative measure of the dirtiness of a firm's imports based on the World Bank's "Industrial Pollution Projection System" (IPPS); results were similar.

Insert Table 5 about here

5.3. Local institutions and global response

In Table 6, we test the hypothesis that U.S. plants facing stronger local institutional pressure will reduce their domestic pollution even more as their parent firm increases imports from LWCs. We expect that plants located in "dirty" counties face more public pressure to reduce toxic emissions than plants located in cleaner counties. We repeated regressions based on Equation 1 while controlling for the environmental condition of the county where a plant is located. As explained before, we ranked counties by the total RSEI toxicity-weighted emissions from all plants located there in 1992. We then selected the 100 "dirtiest" counties according to the

ranking; selecting the 25 or 50 dirtiest counties generated similar results. Our results in Table 6 support our hypothesis. Column (3) suggests that when a parent firm increases its share of imports from LWCs by 10 percentage points, a plant in a clean county will reduce its total toxic emissions by 3.5%. In contrast, a plant located in a dirty county will reduce its total toxic emissions by an additional 4.8%, a very significant difference from plants in clean counties. After controlling for these interaction terms, the main effect of LWC Import Share remains mostly significantly negative.

Insert Table 6 about here

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6. Discussion and Conclusions

This paper investigates the relationship between U.S. firms' imports from low-wage countries and toxic emissions by their domestic plants. Our empirical results suggest that plants release less toxic emission on American soil when their parent firm imports more from LWCs. These effects are stronger for plants located in dirtier U.S. counties, where institutional pressure for pollution reduction is expected to be greater. In addition, goods imported by U.S. firms from LWCs are in more pollution-intensive industries than goods imported from the rest of the world, and U.S. plants produce goods in less-pollution-intensive industries when their parent firm imports more from LWCs. These results provide the first large-sample empirical evidence that U.S. firms offshore both production and pollution to the developing world.

This paper's main contribution is to bring a global aspect to the literature on environmental strategy. While prior literature has examined various environmental strategies such as investment in waste treatment and prevention, product and process innovation, and greenwashing, the current paper puts environmental strategies in a global setting. Globalization relaxes the constraint of local institutions and broadens firms' strategic choices. It allows firms to not only circumvent institutional constraints but also to take advantage of institutional variations and readjust their business portfolios across institutions. In addition, this paper provides the first micro-level empirical evidence of "pollution offshoring" at the plant level and lends further

credence to policy makers' assertions that there should more coordination between international trade agreements and domestic environmental regulations (Keller and Levinson 2002).

The paper's has a few limitations that suggest opportunities for future study. First, while we find evidence that the "green shift" of U.S. manufacturing corresponds with a "brown shift" of imports from LWCs, we do not directly test the impact of globalization on LWCs. On the one hand, offshoring by U.S. firms might bring to LWCs more advanced environmental technologies relative to what LWCs would have used without globalization, thereby causing less pollution for the same size of production in LWCs. On the other hand, offshoring by U.S. firms in more pollution-intensive industries can increase the size of pollution-intensive production in LWCs, exacerbating the pollution problem. A more comprehensive analysis of the impact of globalization on LWC environment can be pursued when such data becomes available.

Our paper also has significant policy implications. It implies that strict domestic environmental standards could mitigate negative externality in one institution but exacerbate negative externality in far-flung institutions due to globalization. With the recent public debates around re- or near-shoring vs. off-shoring, the impact of institutional constraints on the configuration of global production networks becomes increasingly salient. While conventional wisdom assumes that pollution is local, recent research proves it is becoming a global concern. Our empirical findings lend further credence to policy makers' assertions that there should be more coordination between international trade agreements and domestic environmental regulations (Keller and Levinson 2002).

In addition, because the paper analyses offshoring in general and does not differentiate between in-house offshoring and global outsourcing, its theoretical predictions can be ambiguous. On the one hand, offshoring pollution-intensive production to firms' own subsidiaries in LWCs may create additional burdens to manage the potential environmental liabilities and reputational damage. On the other hand, offshoring pollution-intensive production to third-party suppliers does not seem to sufficiently insulate firms from the potential liabilities and reputational damage in cases where something goes wrong, as we have seen in the cases of Nike and Apple. Research on global production networks has suggested that firms should extend the reach of their manufacturing strategy beyond firm boundaries and consider the entire industrial network, including alternative supply relationships such as long-term suppliers, alliance partners, and short-term contractors (Ferdows 2014).

As a supplementary empirical analysis, we examined the imports-pollution relationship for multinational corporations (MNCs), a favorite target for those who criticize globalization. Multinational enterprises mediate a substantial majority of U.S. trade (Bernard, Jensen, and Schott 2009), but the environmental conduct of multinational enterprises is unclear. On the one hand, multinational enterprises often maximize profits by arbitraging regulatory differences across countries, including by circumventing trade barriers through foreign direct investment and internal sourcing (Vernon 1971). If they also arbitrage in differences in environmental regulations across countries, we would expect their imports from LWCs to have a larger negative impact on emissions in the United States. On the other hand, multinational corporations may self-regulate in accordance with their internal standards, including self-regulating their environmental conduct in countries with weak environmental regulations (Christmann and Taylor 2001, Dowell, et al. 2000). In order to shed light on these two opposite expectations, we exploited differences between imports from related parties (foreign subsidiaries or affiliates of MNCs) and imports from independent third parties. Imports are categorized in the LFTTD database as being from related parties if the importer owns, controls, or holds voting power equivalent to at least 6% of the outstanding voting stock or shares of the exporter. Our results (not presented here due to space limitations) suggest that MNCs' imports from related parties in LWCs do not have a statistically different effect on emissions by their U.S. plants, relative to firms' imports from independent parties in LWCs. On average, in our sample, the imports from related parties in LWCs accounted for less than 1% of a firm's total imports. Therefore, the economic significance of importing from related parties in LWCs remains small. In sum, our findings suggest that the environmental effects of importing from LWCs are primarily driven by imports from arm's length transactions with unrelated parties. It will be interesting to explore other dimensions of firm heterogeneity in future studies when more detailed sourcing data becomes available.

Furthermore, our results only speak for the manufacturing sector, which accounts for about one fifth of total U.S. pollution (Levinson, 2009). When data becomes available, future studies could cover broader sectors and investigate further the relationship between global sourcing and environmental performance, particularly as more and more U.S. firms shift to a business model of "servicizing" rather than manufacturing. Finally, data limitations restrained us to the study of trade and pollution substitution by surviving firms. Given that Bernard, Jensen, and Schott (2006)

have shown that importing from LWCs forces many less productive plants to exit, our estimates of pollution benefits of substitution might be underestimated.

In sum, this paper highlights the role of firms' offshoring strategy in improving environmental performance in the United States. In addition to the theoretical contributions highlighted herein, the paper provides empirical evidence at the plant level of offshoring and pollution using a unique data set of a large sample of U.S. firms and plants. It will, we hope, encourage more empirical studies to complement both the extensive efforts in the literature on global manufacturing and environmental strategies and the heated policy debates on the sustainability of globalization.

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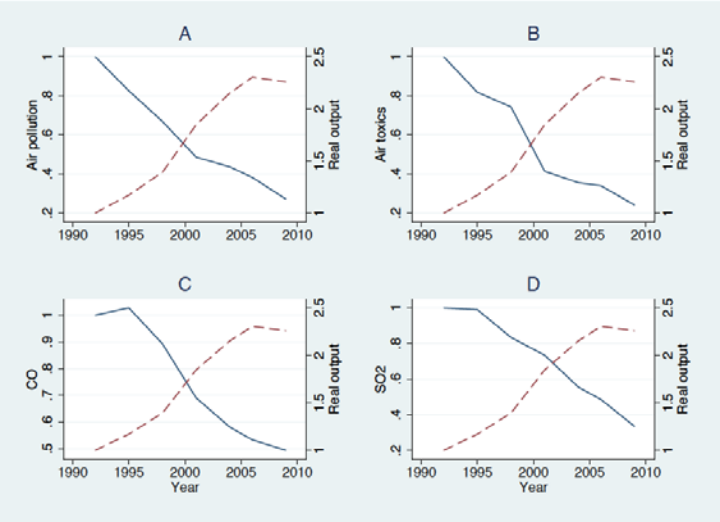
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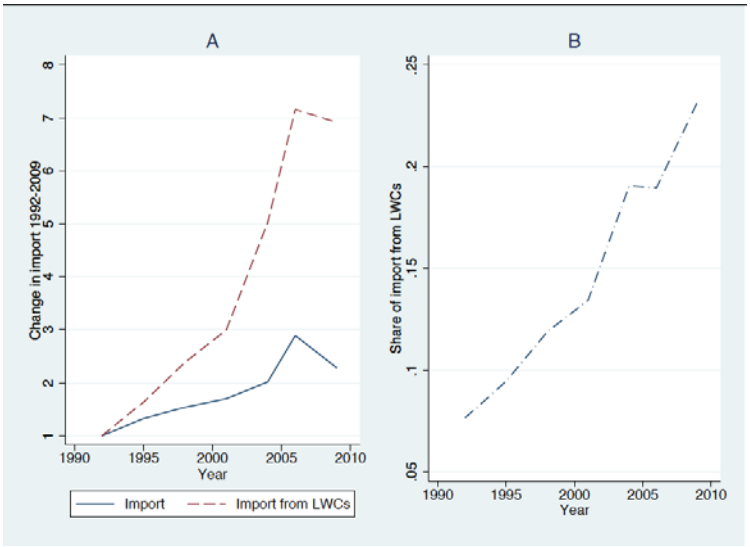
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Figure 1. Pollution and output from U.S. manufacturing, 1992–2009.



Notes. This figure shows air pollution (solid line) and real output (dashed line) from the U.S. manufacturing sector in 1992–2009, where we normalize the 1992 value to be 1. (*A*) Total release of fugitive and stack air from all manufacturing facilities in TRI. (*B*) Total release of toxic content in fugitive and stack air from all manufacturing facilities in TRI. (*C*) Emission of CO from industrial activities in National Emissions Inventory. (*D*) Emission of SO2 from industrial activities in National Emissions Inventory.

Figure 2. U.S. imports and imports from LWCS, 1992–2009.



Notes. Figure 2A shows total U.S. imports and imports from LWCs in 1992–2009, where we normalize the 1992 value of imports to be 1. Figure 2B plots the share of imports originating from LWCs in 1992-2009.

Figure 3. Changes in imports from LWCs and changes in toxic air emissions, 1992–2009.



Notes. This figure shows the changes in each industry's toxic air emissions in 1992–2009 against changes in the share of imports from LWCs. The 1992 value for toxic air emissions is normalized to be 1. (A) Based on pounds of emissions. (B) Based on toxicity weighted pounds of emissions.

Table 1. List of low-wage countries

Afghanistan	China	India	Pakistan
Albania	Comoros	Kenya	Rwanda
Angola	Congo	Lao PDR	Samoa
Armenia	Equatorial Guinea	Lesotho	Sao Tome
Azerbaijan	Eritrea	Madagascar	Sierra Leone
Bangladesh	Ethiopia	Malawi	Somalia
Benin	Gambia	Maldives	Sri Lanka
Bhutan	Georgia	Mali	St. Vincent
Burkina Faso	Ghana	Mauritania	Sudan
Burundi	Guinea	Moldova	Togo
Cambodia	Guinea-Bissau	Mozambique	Uganda
Central African Republic	Guyana	Nepal	Vietnam
Chad	Haiti	Niger	Yemen

Table 2. Summary statistics (main sample)

		N	Mean	SD
(1)	Ln(Toxic emissions)	136574	12.98	6.09
(2)	LWC import share (Parent firm's share of imports from LWCs)	136574	0.16	0.19
(3)	Plant's total value of shipment (in million dollars)	136574	175	586
(4)	Skill intensity	136574	0.35	0.19
(5)	Total capital expenditures (in million dollars)	136574	6.13	36.0
(6)	Parent firm's total import (in million dollars)	136574	711	3200

Table 3. Correlation matrix

		(1)	(2)	(3)	(4)	(5)	(6)
(1)	Ln(Toxic emissions)	1.00					
(2)	LWC import share	-0.07*	1.00				
(3)	Plant's TVS (in million dollars)	0.18*	-0.06*	1.00			
(4)	Skill intensity	-0.09*	0.01*	-0.10*	1.00		
(5)	Total capital expenditures (in million dollars)	0.18*	-0.08*	0.65*	-0.03*	1.00	
(6)	Parent firm's total import (in million dollars)	0.08*	-0.03*	0.34*	-0.03*	0.25*	1.00

*p<0.01.

Table 4. Firms' imports from LWCs and their U.S. plants' toxic emissions

	Toxic Emission		
	(1)	(2)	(3)
LWC Imp	-0.987***	-0.559***	-0.394***
	[0.219]	[0.143]	[0.115]
Ln(Firm Sale)		0.0705**	0.0333*
		[0.031]	[0.020]
Ln(TVS)		0.433***	0.334***
		[0.036]	[0.024]
Skill		-0.773***	-0.598***
		[0.154]	[0.106]
Ln(TCE)		0.0557***	0.0370***
		[0.009]	[0.006]
Ln(Imp)		0.00197	-0.013
		[0.014]	[0.009]
Constant	13.60***	8.345***	4.501***
	[0.065]	[0.440]	[0.307]
Plant FE			Yes
Industry FE	Yes	Yes	
Year FE	Yes	Yes	Yes
Observations	136574	136574	136574
R-squared	0.194	0.712	0.782

N=136,574. This table reports regression estimates of the correlation between firms' imports from LWCs and their plant-level toxic emissions in the U.S. in 1992–2009, based on Equation 1. The sample includes all firms that import and are surveyed by the TRI. Standard errors are clustered at the firm level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 5. Firms' imports from LWCs and their pollution contents

	Ln(Toxic Import)		Toxic Imp/Imp	
	(1)	(2)	(3)	(4)
LWC Imp	0.647***	0.369***	2.167**	0.590**
	[0.091]	[0.055]	[1.064]	[0.286]
Ln(Imports)	1.449***	1.179***	0.156	0.252***
	[0.014]	[0.008]	[0.310]	[0.056]
Ln(Firm Sales)	0.0651***	0.0690***	0.0924	-0.0668
	[0.024]	[0.014]	[0.186]	[0.073]
Constant	5.727***	3.869***	7.900***	3.778***
	[0.259]	[0.150]	[2.705]	[0.899]
Firm FE		Yes		Yes
Industry FE	Yes		Yes	
Year FE	Yes	Yes	Yes	Yes
Observations	277768	277768	277768	277768
Adjusted R	0.679	0.789	0.08	0.65

N=277,768. This table reports regression estimates of the correlation between firms' imports from LWCs and the pollution intensity of their imports, from 1992 to 2009, based on Eq. 2. The sample includes all firms that import. Pollution intensity is calculated based on the TRI. Standard errors are clustered at the firm level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 6. Firms' imports from LWCs and their U.S. plants' toxic emissions, heterogeneity across counties

	Toxic Emission		
	(1)	(2)	(3)
LWC Imp	-0.981***	-0.374***	-0.483***
	[0.219]	[0.096]	[0.090]
Dirty county	0.832***	0.827***	0.568***
	[0.090]	[0.041]	[0.048]
LWC Imp*Dirty county		-0.565***	-0.357**
		[0.215]	[0.176]
Ln(Firm Sale)		0.0831***	0.0704***
		[0.014]	[0.019]
Ln(TVS)		0.767***	0.430***
		[0.019]	[0.026]
Skill		-1.744***	-0.774***
		[0.093]	[0.105]
Ln(TCE)		0.180***	0.0554***
		[0.010]	[0.007]
Ln(Imp)		-0.0393***	0.00201
		[0.008]	[0.008]
Constant	13.38***	4.414***	8.234***
	[0.068]	[0.181]	[0.311]
Industry FE		Yes	
Year FE	Yes	Yes	Yes
Plant FE			Yes
Observations	136574	136574	136574
Adjusted R-squared	0.197	0.233	0.713

N=136,574. This table reports regression estimates of the correlation between firms' imports from LWCs and their plant-level toxic emissions in the U.S. in 1992–2009, based on Equation 1. The sample includes all firms that import and are surveyed by the TRI. Standard errors are clustered at the firm level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.